

KUSH MEHTA

B.Tech Electronics & Communication Engineering | Minor: Data Science

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EDUCATION

Institute of Technology, Nirma University

Ahmedabad, India

B.Tech, Electronics & Communication Engineering — Minor: Data Science | **CGPA: 8.43 / 10** Aug 2023 – May 2027

- Relevant Coursework: Embedded Systems, Computer Architecture, VLSI Design, Digital Signal Processing, Image Processing, Signals & Systems, Research Methodology & Seminar, Communication Systems, Scripting Languages
- Minor in Data Science — coursework in statistics, data analysis & visualization, machine learning, Analytics of IoT

LEADERSHIP & EXPERIENCE

Electronics Lead — Team Arrow UAV Club

Aug 2023 – Present

Nirma University Official Aeromodelling & Autonomous Systems Club

Ahmedabad, India

- Lead all electronics systems design and integration for competition-grade fixed-wing aircraft across 4 national and international competitions
- Designed autonomous systems, flight controller configuration (ArduPilot/Pixhawk), RC telemetry links, onboard payload electronics, and ESC/motor selection for thrust optimization
- Competed at SAE Aero Design West 2025 (USA), SAE Aero Design East 2026 (USA), SAEISS DDC 2024,2025,2026 (India), and Technoxian 2024 (India) — achieving multiple top-5 podium finishes
- Led a team of 8–12 members; mentored junior members on sensor fusion, firmware development, and avionics integration
- Coordinated cross-functional collaboration between aerodynamics, structures, and electronics sub-teams for design integration reviews

PROJECTS

[LINK](#)

Secure GPS Location Tracker — ESP32 + AES-128 + MQTT + Google Maps

Mar 2026

IoT & Embedded Security | Nirma University

- Designed end-to-end IoT security pipeline: ESP32 reads GPS coordinates via TinyGPS++ over UART2, serializes to JSON, encrypts with AES-128 ECB (mbedtls built into ESP32 SDK)
- Published hex-encoded ciphertext to HiveMQ MQTT broker; Python subscriber (paho-mqtt + PyCryptodome) decrypts and automatically opens Google Maps at the live location
- Validated at Nirma University campus — Serial Monitor confirmed GPS fix (23.13°N, 72.54°E), Mosquitto terminal showed continuous encrypted stream, browser opened correct coordinates
- Technologies: ESP32, TinyGPS++, mbedtls AES-128, MQTT/HiveMQ, Python, paho-mqtt, PyCryptodome

IoT Cryptography Benchmark — AES-GCM vs ASCON vs DES vs ECC P-256

Jan 2026

Research Project | Nirma University

- Benchmarked 4 cryptographic algorithms across 5,000 realistic IoT JSON payloads (64-byte fixed size) simulating temperature/voltage sensor data
- ASCON-128 loaded as compiled C DLL via Python ctypes (native speed); peak memory traced with tracemalloc for DES, AES-GCM, and ECC
- Results: ASCON 78× faster than AES-GCM and 12× faster than DES — validating ASCON's design intent for low-power edge devices; ECC handshake completed in 8.45ms
- Technologies: Python, ctypes, PyCryptodome, tracemalloc, Matplotlib, C (ASCON reference implementation)

gem5 L1 Cache Sensitivity Analysis — Matrix Multiply & AES Workloads

Feb 2026

Computer Architecture | Nirma University

- Conducted 3-experiment study on gem5 v25.1 with X86 TimingSimpleCPU: cache capacity (4kB–64kB), workload sensitivity (matrix multiply vs AES SubBytes), and associativity (1-way to 8-way)
- Matrix multiply: miss rate dropped from 5.06% to 0.04% and IPC improved from 0.277 to 0.310 as cache grew 4kB→64kB; AES showed flat 0.14% miss rate — bottleneck in execution units not cache
- Associativity experiment: 1-way→8-way reduced miss rate from 4.74% to 0.34%, IPC gain +10.4%; concluded highly-associative modest-size caches optimal for UAV/IoT applications

Real-Time Virtual Try-On Sunglasses — Classical Computer Vision

Sep 2025

Computer Vision & AR | Nirma University

- Built real-time AR try-on using classical CV — no deep learning or GPU required; pipeline: CLAHE local contrast enhancement → Viola-Jones Haar Cascade (face + eye detection) → geometric pose estimation → overlay
- Glasses scaled to inter-pupillary distance ($\times 2.3$), rotated via atan2 on eye centers; EMA smoothing ($\alpha=0.2$) applied independently to x, y, scale, and rotation to eliminate jitter
- Color customization via HSV hue shifting (mod 180, preserves alpha channel); per-pixel alpha blending for compositing; supports multiple glasses models (n-key cycle)
- Technologies: Python, OpenCV, NumPy

Investment Portfolio Analyzer

Oct 2025

Data Science & OOP | Course: PSC, Semester 5, Nirma University

- Designed OOP-based portfolio analysis system: Investment base class with 5 subclasses (Stock, RealEstate, BankDeposit, MutualFund, DebtBond) using Python inheritance
- Processed 50-asset portfolio CSV with Pandas groupby aggregation; portfolio metrics: ₹92.1M current value, ₹100.1M predicted end-of-FY, 9% average growth rate
- Best performer: Stock #21 (+39.83% growth); Worst: Stock #8 (-21.16%); visualized with pie chart (composition) and bar chart (growth by asset class); exported summary to report.txt

Log Analytics & Anomaly Detection Tool

Mar 2026

Embedded Systems & Linux | Course: 3EC504ME24, Nirma University

- Two-module architecture: collector.sh (Bash) samples CPU, memory, disk, and failed login metrics on configurable intervals; detector.py (Python) applies $\mu+2\sigma$ statistical thresholding for anomaly flagging
- Validated via stress test: yes > /dev/null background processes triggered CPU threshold breach; known cold-start limitation requires baseline data before thresholds become accurate

IPL Match Prediction — KNN Classifier

Apr 2024

Introduction to Machine Learning | Course: 1CS101, Semester II, Nirma University

- First ML project: KNN classifier ($K=9$, Euclidean distance) trained on 636 IPL matches using 4 features (team names, toss winner, toss decision); achieved 57.81% accuracy
- Strong performers: Mumbai Indians (0.64 precision/0.74 recall), Kolkata Knight Riders (0.65/0.73); accuracy reflects cricket's inherent unpredictability with limited feature set

RESEARCH

Simulation & Comparative Analysis of Lightweight Cryptography for Embedded Systems 2025–2026

Research Methodology & Seminar — Nirma University | Status: In Progress

- Systematic study comparing AES-128-GCM, DES-CBC, ECC P-256, and ASCON-128 on throughput, peak memory, and energy efficiency across 5,000 IoT payloads at multiple sizes (32B, 64B, 128B)
- ASCON loaded via compiled C DLL (ctypes) for native-speed benchmarking; Python tracemalloc for memory profiling; statistical significance tested across 10 trial runs
- Findings support ASCON-128 as the NIST-recommended lightweight standard for resource-constrained edge devices; results submitted as course seminar report

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Verilog HDL, Bash, MATLAB, SQL, Assembly Language, TCL, MATLAB

Embedded & Hardware: ESP32, Raspberry Pi, Arduino, FPGA, Keil uVision, mbedTLS, ArduPilot/Mission Planner, Pixhawk Flight Controllers, 8051

IoT & Communication Protocols: MQTT (HiveMQ, Mosquitto), Node-RED, TinyGPS++, UART, I2C, SPI, Wi-Fi, BLE

Cryptography: AES-128 (ECB/GCM), DES, ECC P-256 (ECDH/HKDF), ASCON-128, PyCryptodome, ctypes DLL interface

Data Science & Analytics: Pandas, NumPy, Matplotlib, Scikit-learn, Tableau, KNIME, Statistical Analysis

Design & Simulation: SolidWorks, Fusion 360, XFLR5, gem5 Simulator, MICROWIND, Quartus II, Simulink

Tools & Platforms: Git/GitHub, Linux/Ubuntu, Arduino IDE, Modelsim, Solidworks

COMPETITIONS & ACHIEVEMENTS

- SAE Aero Design West 2025 (Los Angeles, CA, USA) — 5th internationally (Mission); Electronics Lead
- SAEISS DDC 2024 (India) — AIR 2; Electronics Lead
- SAEISS DDC 2025 (India) — AIR 3; Electronics Lead